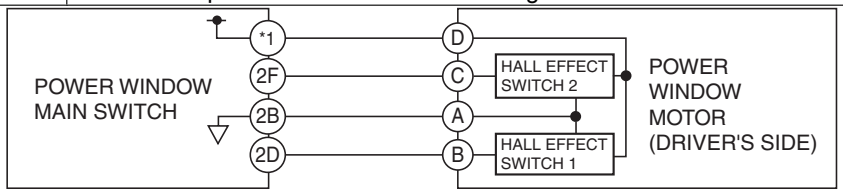
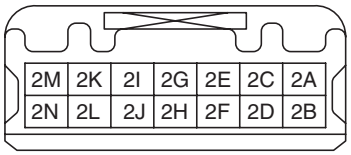
DRIVER-SIDE DOOR GLASS CAN BE OPERATED MANUALLY BUT CANNOT BE OPERATED AUTOMATICALLY [POWER WINDOW SYSTEM]

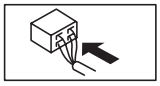
id0903a0479800

Description	Driver-side door glass cannot be operated automatically but can be operated manually using power window main switch
Possible Causes	• Power window main switch malfunction • Power window motor malfunction • Open circuit in wiring harness between the following terminals: — Power window main switch terminal 2D and power window motor terminal B — Power window main switch terminal 2B and power window motor terminal A — Power window main switch terminal 2F and power window motor terminal C — Power window main switch terminal 2N (with door lock switch)/2M (without door lock switch) and power window motor terminal D • Short to ground in wiring harness between the following terminals: — Power window main switch terminal 2D and power window motor terminal B — Power window main switch terminal 2F and power window motor terminal C — Power window main switch terminal 2N (with door lock switch)/2M (without door lock switch) and power window motor terminal D • Short to power supply in wiring harness between the following terminals: — Power window main switch terminal 2D and power window motor terminal B — Power window main switch terminal 2F and power window motor terminal C • Connector poor contact or terminal damage

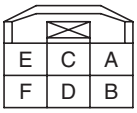


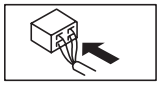
POWER WINDOW MAIN SWITCH WIRING HARNESS-SIDE CONNECTOR





POWER WINDOW MOTOR (LF) WIRING HARNESS-SIDE CONNECTOR





	*1
WITHOUT DOOR LOCK SWITCH	2M
WITH DOOR LOCK SWITCH	2N

Diagnostic Procedure

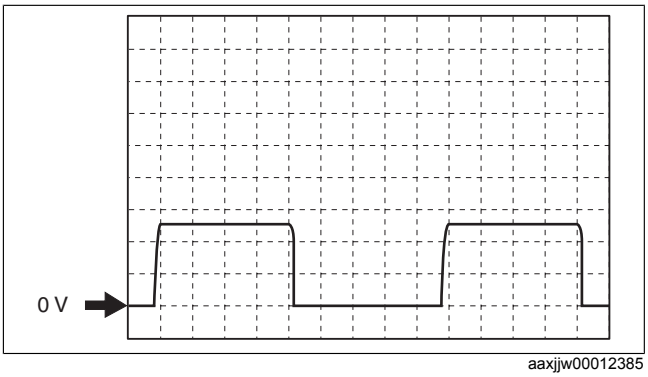
Step	Inspection		Action
1	**VERIFY IF MALFUNCTION CAUSE IS OPERATION UNDER OPERATION PERMISSION CONDITION** • Perform the power window system initialization procedure. • Does the driver-side door glass operate automatically using the power window main switch?	Yes	System is normal. (Tell the customer about power window system initialization procedure.)
		No	Go to the next step.
2	**POWER WINDOW MAIN SWITCH CONNECTOR INSPECTION** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the power window main switch connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 10.

Step	Inspection		Action
3	**POWER WINDOW MOTOR CONNECTOR INSPECTION** • Disconnect the power window motor connector. • Inspect the connector engagement and connection condition and inspect the terminals for damage, deformation, corrosion, or disconnection. • Is the connector normal?	Yes	Go to the next step.
		No	Repair or replace the connector, then go to Step 10.
4	**POWER WINDOW MOTOR TERMINAL D INSPECTION** • Reconnect all the disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Measure the voltage at the power window motor terminal D (vehicle wiring harness side). • Is the voltage B+?	Yes	Go to Step 6.
		No	Go to the next step.
5	**POWER WINDOW MAIN SWITCH TERMINAL 2N (WITH DOOR LOCK SWITCH)/2M (WITHOUT DOOR LOCK SWITCH) INSPECTION** • Measure the voltage at power window main switch terminal 2N (with door lock switch)/2M (without door lock switch) (vehicle wiring harness side). • Is the voltage B+?	Yes	Refer to the wiring diagram and verify if there is a common connector between power window main switch terminal 2N (with door lock switch)/2M (without door lock switch) and power window motor terminal D. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open or short circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open or short circuit. Go to Step 10.
		No	Replace the power window main switch, then go to Step 10. (See POWER WINDOW MAIN SWITCH REMOVAL/INSTALLATION.)
6	**DETERMINE IF MALFUNCTION CAUSE IS POWER WINDOW MAIN SWITCH INTERNAL GROUND SYSTEM** • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Inspect for continuity between power window motor terminal A (vehicle wiring harness side) and body ground. • Is there continuity?	Yes	Go to Step 8.
		No	Go to the next step.
7	**VERIFY IF MALFUNCTION CAUSE IS OPEN CIRCUIT IN WIRING HARNESS BETWEEN POWER WINDOW MAIN SWITCH AND POWER WINDOW MOTOR** • Disconnect the power window main switch connector and power window motor connector. • Inspect for continuity between the following terminals (vehicle wiring harness). — Power window main switch terminal 2B and power window motor terminal A • Is there continuity?	Yes	Replace the power window main switch, then go to Step 10. (See POWER WINDOW MAIN SWITCH REMOVAL/INSTALLATION.)
		No	Refer to the wiring diagram and verify if there is a common connector between power window main switch terminal 2B and power window motor terminal A. **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 10.

Step	Inspection		Action
8	**POWER WINDOW MAIN SWITCH TERMINALS 2F AND 2D INSPECTION** • Reconnect all the disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Operate the driver-side power window to measure the pulse at power window main switch terminals 2F and 2D. • Measure the pulse at power window main switch terminals 2F and 2D. (See Inspection Using an Oscilloscope.) • Is the pulse normal?	Yes	**Note** • The power window motor is normal because the pulse sent from the power window motor to the power window main switch is normal. Replace the power window main switch, then go to Step 10. (See POWER WINDOW MAIN SWITCH REMOVAL/INSTALLATION.)
		No	Go to the next step.
9	**POWER WINDOW MOTOR TERMINALS B AND C INSPECTION** • Operate the driver-side power window to measure the pulse at power window motor terminals B and C. (See Inspection Using an Oscilloscope.) • Is the pulse normal?	Yes	Refer to the wiring diagram and verify if there is a common connector between the following terminals. • Power window main switch terminal 2D and power window motor terminal B • Power window main switch terminal 2F and power window motor terminal C **If there is a common connector:** • Inspect the common connector and terminals for corrosion, damage, or disconnection and the common wiring harnesses for an open or short circuit to determine the malfunctioning location. • Repair or replace the malfunctioning location. **If there is no common connector:** • Repair or replace the wiring harness which has an open or short circuit. Go to the next step.
		No	Replace the power window motor, then go to the next step. (See FRONT POWER WINDOW MOTOR REMOVAL/INSTALLATION.)
10	**VERIFY IF MALFUNCTION CAUSE IS CORRECTED** • Does the power window system operate normally?	Yes	Troubleshooting completed. (Explain the problem to the customer.)
		No	Verify the malfunction symptom in the symptom troubleshooting chart and perform the other applicable malfunction diagnosis.

Inspection Using an Oscilloscope

- Power window main switch: 2D (+), 2F (+) ↔ ground (-)
- Power window motor: B (+), C (+) ↔ ground (-)
- Oscilloscope setting: 2 V/DIV (Y), 1 ms/DIV (X), DC range



aaxjjw00012385